Calculation Engine

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Chapter 1

Calculation Engine Project

This project is a small multi-file C++ program. It contains:

- Logger.h: a simple class that prints log messages.
- CalculationEngine.h: a class for math operations that uses the Logger.
- main.cpp: the application entry point.

The documentation is generated using Doxygen.

Chapter 2

Todo List

Member CalculationEngine::power_calculation (double base, int exponent)

Add support for negative exponents.

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

CalculationEngine
: Performs simple mathematical operations 9

Logger
: A simple class that prints messages to the console 10

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

CalculationEngine.h
: Defines a class for mathematical operations 13

Logger.h
: Defines a simple Logging class 14

main.cpp
: Application entry point 16

Chapter 5

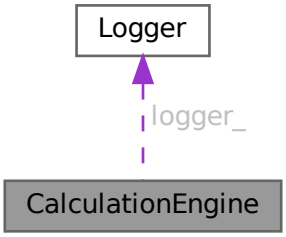
Class Documentation

5.1 CalculationEngine Class Reference

Performs simple mathematical operations.

`#include <CalculationEngine.h>`

Collaboration diagram for CalculationEngine:



Public Member Functions

- double power_calculation (double base, int exponent)

  Calculates base raised to the power of exponent.

Private Attributes

- Logger logger_ {"CalcEngine"}

5.1.1 Detailed Description

Performs simple mathematical operations.

Every operation writes a message using the Logger class, so the user can see which calculations were performed.

5.1.2 Member Function Documentation

5.1.2.1 power_calculation()

```
double CalculationEngine::power_calculation (
            double base,
            int exponent )  [inline]
```

Calculates base raised to the power of exponent.

Parameters

base	The base number.
exponent	The exponent (a non-negative whole number).

Returns

The value of base raised to the power exponent.

Todo Add support for negative exponents.

5.1.3 Member Data Documentation

5.1.3.1 logger_

```
Logger CalculationEngine::logger_ {"CalcEngine"}  [private]
```

Logger used by this class.

The documentation for this class was generated from the following file:

- CalculationEngine.h

5.2 Logger Class Reference

A simple class that prints messages to the console.

```
#include <Logger.h>
```

Public Member Functions

- Logger (const std::string &name, LogLevel level=INFO)
  Constructs a Logger.
- void log (const std::string &message) const
  Prints a message to the console.

Private Attributes

- std::string name_
- LogLevel level_

5.2.1 Detailed Description

A simple class that prints messages to the console.

5.2.2 Constructor & Destructor Documentation

5.2.2.1 Logger()

```
Logger::Logger (
            const std::string & name,
            LogLevel level = INFO )  [inline]
```

Constructs a Logger.

Parameters

name	Name printed at the start of every message.
level	Level used when log() is called without a level.

5.2.3 Member Function Documentation

5.2.3.1 log()

```
void Logger::log (
            const std::string & message ) const  [inline]
```

Prints a message to the console.

Parameters

message	The text to print.

5.2.4 Member Data Documentation

5.2.4.1 level_

```
LogLevel Logger::level_  [private]
```

Default level of the logger.

5.2.4.2 name_

```
std::string Logger::name_  [private]
```

Name of the logger.

The documentation for this class was generated from the following file:

- Logger.h

Chapter 6

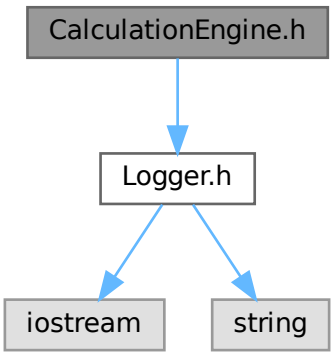
File Documentation

6.1 CalculationEngine.h File Reference

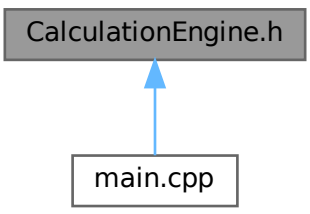
Defines a class for mathematical operations.

```
#include "Logger.h"
```

Include dependency graph for CalculationEngine.h:



This graph shows which files directly or indirectly include this file:



Classes

- class CalculationEngine

  Performs simple mathematical operations.

6.1.1 Detailed Description

Defines a class for mathematical operations.

6.2 CalculationEngine.h

Go to the documentation of this file.

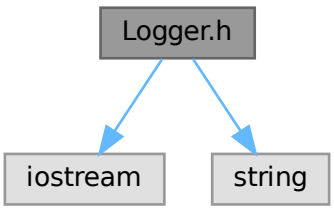
```
00001
00005 #ifndef CALCULATIONENGINE_H
00006 #define CALCULATIONENGINE_H
00007
00008 #include "Logger.h"
00009
00017 class CalculationEngine {
00018 public:
00028     double power_calculation(double base, int exponent) {
00029         logger_.log("Calculating power");
00030         double result = 1.0;
00031         for (int i = 0; i < exponent; i++) {
00032             result *= base;
00033         }
00034         return result;
00035     }
00036
00037 private:
00038     Logger logger_{"CalcEngine"};
00039 };
00040
00041 #endif
```

6.3 Logger.h File Reference

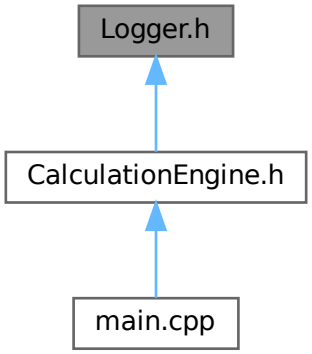
Defines a simple Logging class.

```
#include <iostream>
#include <string>
```

Include dependency graph for Logger.h:



This graph shows which files directly or indirectly include this file:



Classes

- class Logger

  A simple class that prints messages to the console.

Enumerations

- enum LogLevel { INFO , WARNING , ERROR }

  Severity level of a log message.

6.3.1 Detailed Description

Defines a simple Logging class.

6.3.2 Enumeration Type Documentation

6.3.2.1 LogLevel

```
enum LogLevel
```

Severity level of a log message.

Enumerator

INFO	General information message.
WARNING	Something unexpected, but the program can continue.
ERROR	A serious problem occurred.

6.4 Logger.h

Go to the documentation of this file.

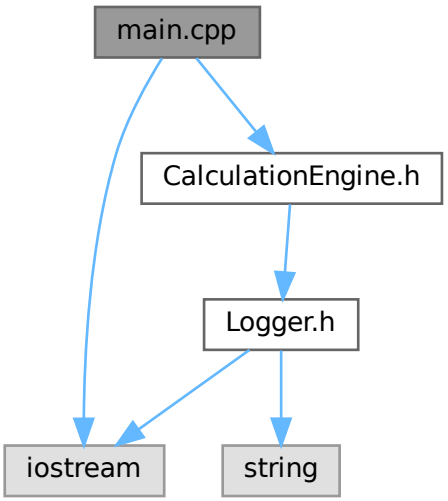
```
00001
00005 #ifndef LOGGER_H
00006 #define LOGGER_H
00007
00008 #include <iostream>
00009 #include <string>
00010
00026 enum LogLevel {
00027     INFO,
00028     WARNING,
00029     ERROR
00030 };
00031
00036 class Logger {
00037 public:
00043     Logger(const std::string& name, LogLevel level = INFO)
00044         : name_(name), level_(level) {}
00045
00050     void log(const std::string& message) const {
00051         const char* labels[] = {"INFO", "WARNING", "ERROR"};
00052         std::cout « "[" « name_ « "] " « labels[level_] « ": "
00053                   « message « std::endl;
00054     }
00055
00056 private:
00057     std::string name_;
00058     LogLevel level_;
00059 };
00060
00061 #endif
```

6.5 main.cpp File Reference

Application entry point.

```
#include <iostream>
#include "CalculationEngine.h"
```

Include dependency graph for main.cpp:



Functions

- int main ()

  Runs a sample calculation.

6.5.1 Detailed Description

Application entry point.

6.5.2 Function Documentation

6.5.2.1 main()

```
int main ( )
```

Runs a sample calculation.

Returns

0 on success.

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